

Requirements Specification and Analysis I

What are Requirements?

- A Requirement is:
 - A capability or condition required from the system.
- What is involved in requirements analysis and specification?
 - Determine what is expected by the client from the system. (Gather and Analyze)
 - Document those in a form that is clear to the client as well as to the development team members. (Document)

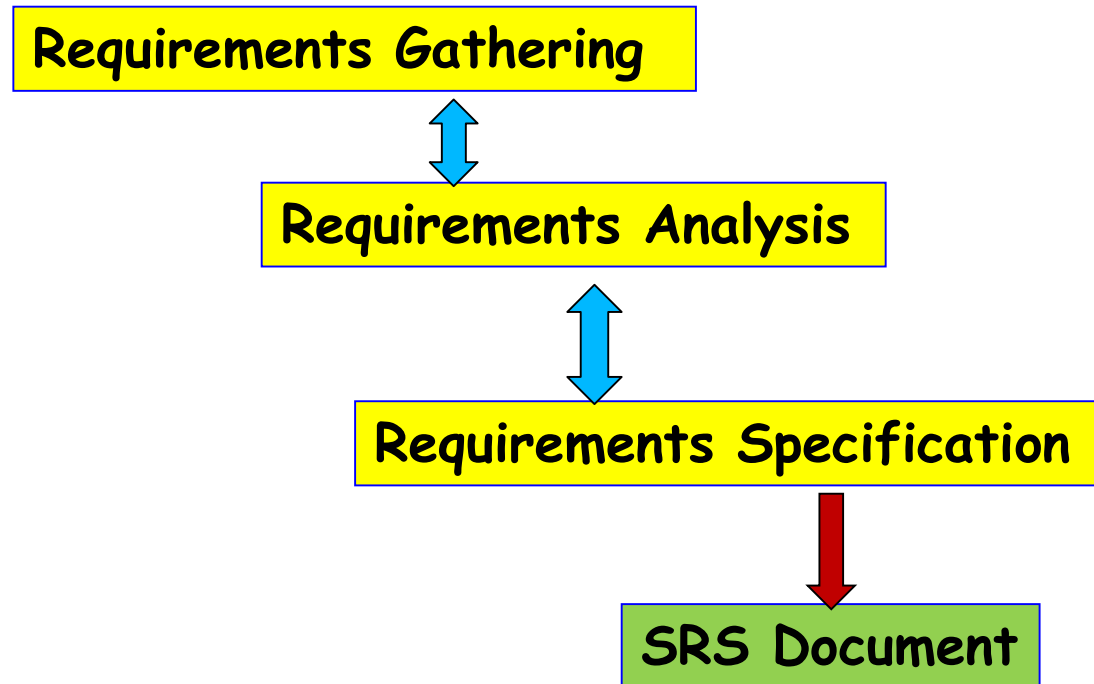
Understanding and specifying requirements

- **For toy problems:** understanding and specifying requirements is rather easy...
- **For industry-standard problems:** Probably the hardest, most problematic and error prone among development tasks...
- The task of requirements specification :
 - **Input:** User needs that are hopefully fully understood by the users.
 - **Output:** Precise statement of what the software will do.

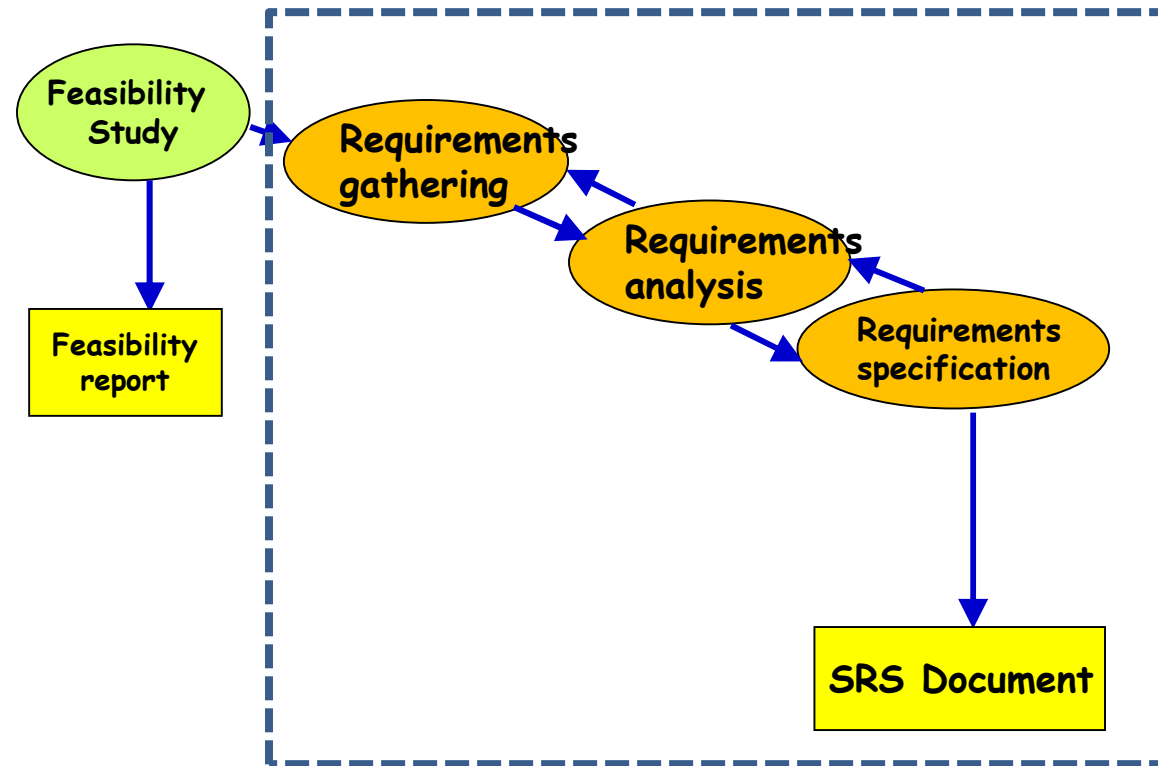
Requirements for Products

- When a company plans to develop a generic product:
 - Who gives the requirements?
- **The sales personnel!**

Activities in Requirements Analysis and Specification



Requirements Engineering Process



Requirements Analysis and Specification

- Requirements Gathering:
 - Fully understand the user requirements.
- Requirements Analysis:
 - Remove inconsistencies, anomalies, etc. from requirements.
- Requirements Specification:
 - Document requirements properly in an SRS document.

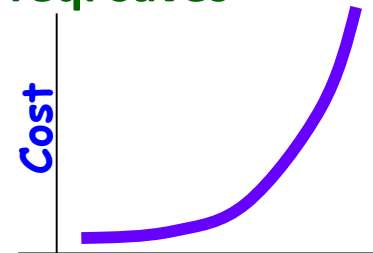
- **Good SRS reduces development cost:**

- Req. errors are expensive to fix later
- Req. changes cost a lot (typically 40% of requirements change later)
- Good SRS can minimize changes and errors
- **Substantial savings --- effort spent during req. saves multiple times that effort**

Need for SRS...

- **An Example:**

- Cost of fixing errors in req. , design , coding , acceptance testing and operation increases exponentially



What are the Uses of an SRS Document?

- Establishes the basis for agreement between the customers and the suppliers
- Forms the starting point for development.
- Provide a basis for estimating costs and schedules.
- Provide a basis for validation and verification.
- Provide a basis for user manual preparation.
- Serves as a basis for later enhancements.

Forms A Basis for User Manual

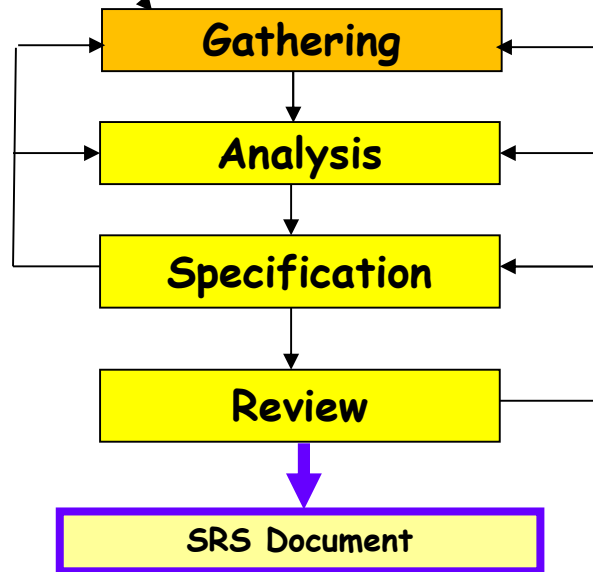
- The SRS serves as the basis for writing User Manual for the software:
 - **User Manual: Describes the functionality from the perspective of a user --- An important document for users.**
 - Typically also describes how to carry out the required tasks with examples.

- SRS intended for a diverse audience:
 - **Customers and users use it for validation, contract, ...**
 - **Systems (requirements) analysts**
 - **Developers, programmers to implement the system**
 - **Testers use it to check whether requirements have been met**
 - **Project Managers to measure and control the project**
- Different levels of detail and formality is needed for each audience
- Different templates for requirements specifications used by companies:
 - Often variations of **IEEE 830**

**SRS Document:
Stakeholders**

User needs

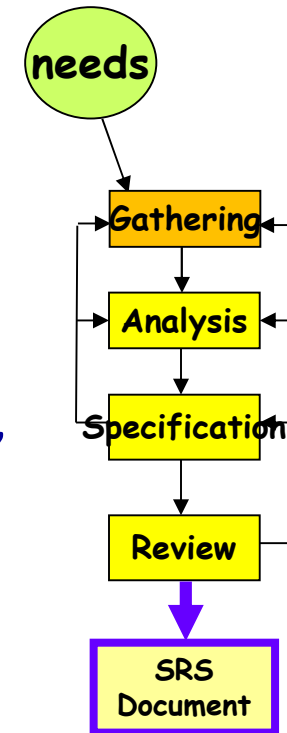
Requirement process..



- Specification and review may lead to further gathering and analysis.

How to Gather Requirements?

- Observe existing (manual) systems,
- Study existing procedures,
- Discuss with customer and end-users,
- Input and Output analysis
- Analyze what needs to be done



Requirements Gathering Activities

- 1. Study existing documentation
- 2. Interview
- 3. Task analysis
- 4. Scenario analysis
- 5. Form analysis

Requirements Gathering (CONT.)

- In the absence of a working system,
 - Lot of imagination and creativity are required.
- Interacting with the customer to gather relevant data:
 - Requires a lot of experience.

Requirements Gathering (CONT.)

- Some desirable attributes of a good requirements analyst:
 - Good interaction skills,
 - Imagination and creativity,
 - Experience...

Case Study: Automation of Office Work at CSE Dept.

- The academic, inventory, and financial information at the CSE department:
 - At present carried through manual processing by two office clerks, a store keeper, and two attendants.
- Considering the low budget he had at his disposal:
 - The HoD entrusted the work to a team of student volunteers.

Case Study: Automation of Office Work at CSE Dept.

- The team was first briefed by the HoD:
 - Concerning the specific activities to be automated.
- The analysts first discussed with the two office clerks:
 - Regarding their specific responsibilities (tasks) that were to be automated.
- The analyst also interviewed student and faculty representatives who would also use the software.

Interview

Case Study: Automation of Office Work at CSE Dept.

- For each task that a user needs the software to perform, they asked:
 - The steps through which these are to be performed.
 - The various scenarios that might arise for each task.
- Also collected the different types of forms that were being used.

Task and Scenario
Analysis

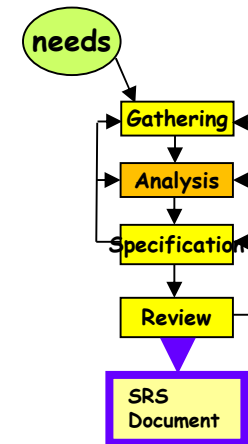
Form
Analysis

Case Study: Automation of Office Work at CSE Dept.

- The analysts understood the requirements for the system from various user groups: **Requirements Analysis**
 - Identified inconsistencies, ambiguities, incompleteness.
- Resolved the requirements problems through discussions with users:
 - Resolved a few issues which the users were unable to resolve through discussion with the HoD.
- Documented the requirements in the form of an SRS document. **Requirements Specification**

Analysis of Gathered Requirements

- Main purpose of req. analysis:
 - Clearly understand user requirements,
 - **Detect inconsistencies, ambiguities, and incompleteness.**
- Incompleteness and inconsistencies:
 - Resolved through further discussions with the end-users and the customers.



Ambiguity

“When temperature becomes high, start cooler”

Do you notice any problems?

- Above what threshold we consider the temperature to be high?

Inconsistent Requirement

- Some part of the requirement:
 - contradicts some other requirement.
- Example:
 - One customer says turn off heater and open water shower when temperature $> 100^{\circ}\text{C}$
 - Another customer says turn off heater and turn ON cooler when temperature $> 100^{\circ}\text{C}$

- Some requirements not included:

- Possibly due to oversight.

**Incomplete
Requirement**

- Example:

- The analyst has not recorded that when temperature falls below 90°C :

- heater should be turned ON
 - water shower turned OFF.

Analysis of the Gathered Requirements

- Requirements analysis involves:
 - Obtaining a clear, in-depth understanding of the software to be developed
 - Remove all ambiguities and inconsistencies from the initial customer perception of the problem.

Analysis of the Gathered Requirements (CONT.)

- It is quite difficult to obtain:
 - A clear, in-depth understanding of the problem:
 - Especially if there is no working model of the problem.

Analysis of the Gathered Requirements

(CONT.)

- Experienced analysts take considerable time:
 - **Clearly understand the exact requirements the customer has in his mind.**

Analysis of the Gathered Requirements

(CONT.)

- Experienced systems analysts know -
often as a result of painful experiences ---
 - **“Without a clear understanding of the problem, it is impossible to develop a satisfactory system.”**

Analysis of the Gathered Requirements

- Several things about the project should be clearly understood:
 - What is the problem?
 - What are the possible solutions to the problem?
 - What complexities might arise while solving the problem?

Analysis of the Gathered Requirements

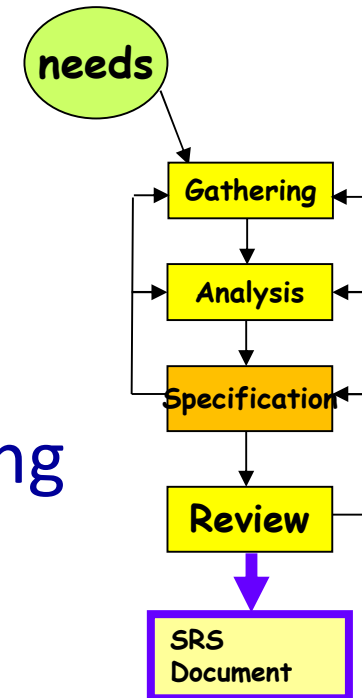
- Some anomalies and inconsistencies can be very subtle:
 - Escape even most experienced eyes.
 - If a formal specification of the system is constructed,
 - Many of the subtle anomalies and inconsistencies get detected.

Analysis of the Gathered Requirements_(CONT.)

- After collecting all data regarding the system to be developed,
 - Remove all inconsistencies and anomalies from the requirements,
 - Systematically organize requirements into a Software Requirements Specification (SRS) document.

Software Requirements Specification

- Main aim:
 - Systematically organize the requirements arrived during requirements analysis.
 - Document requirements properly.

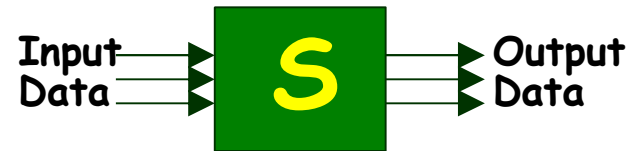


SRS Document

- As already pointed out--- useful in various contexts:
 - **Statement of user needs**
 - **Contract document**
 - **Reference document**
 - **Definition for implementation**

SRS Document (CONT.)

- SRS document is known as **black-box specification**:



- The system is considered as a black box whose internal details are not known.
- Only its visible external (i.e. input/output) behavior is documented.

SRS Document (CONT.)

- SRS document concentrates on:
 - **What** needs to be done in terms of input-output behaviour
 - Carefully avoids the solution (“**how to do**”) aspects.

SRS Document (CONT.)

- The requirements at this stage:
 - Written using end-user terminology.
- If necessary:
 - Later a formal requirement specification may be developed from it.

- **It should be concise**
 - and at the same time should not be ambiguous.
- **It should specify what the system must do**
 - and not say how to do it.
- **Easy to change.,**
 - i.e. it should be well-structured.
- **It should be consistent.**
- **It should be complete.**

**Properties of a Good
SRS Document**

Properties of a Good SRS Document

(cont...)

- **It should be traceable**
 - You should be able to trace which part of the specification corresponds to which part of the design, code, etc and vice versa.
- **It should be verifiable**
 - e.g. “system should be user friendly” is not verifiable

SRS should not include...

- **Project development plans**
 - E.g. cost, staffing, schedules, methods, tools, etc
 - Lifetime of SRS is until the software is made obsolete
 - Lifetime of development plans is much shorter
- **Product assurance plans**
 - Configuration Management, Verification & Validation, test plans, Quality Assurance, etc
 - Different audiences
 - Different lifetimes
- **Designs**
 - Requirements and designs have different audiences
 - Analysis and design are different areas of expertise

Thank You!!